

How do Organisms Reproduce?

Asexual reproduction in single-celled and simple organisms, sexual reproduction in flowering plants, and the human reproductive systems with menstrual cycle.

Introduction

Reproduction is the process by which living organisms produce new individuals of the same kind. It is not needed for the survival of one individual. It is needed for the continuation of the species. Every organism has a limited life span, so reproduction keeps the species going.

In this chapter we study two main types. Asexual reproduction involves a single parent and usually produces very similar offspring, with small variations possible during DNA copying. Sexual reproduction combines genetic material and produces greater variation.

Why Reproduction Needs DNA Copying

DNA is present in the chromosomes inside the nucleus of a cell. It carries all the information for body design. When a cell divides to form new cells, it makes a copy of its DNA.

DNA copying is the basic event in reproduction. The copy is made with the help of chemical reactions. These reactions are not always 100 percent accurate. Small errors can happen. This leads to small differences between parent and offspring. These differences are called variations.

Variations are important. If the environment changes, some variations may help the organism to survive. So DNA copying with small variations is useful for the survival of the species.

Key terms

Variation: differences in characters among individuals of the same species.

DNA (Deoxyribonucleic acid): the molecule in chromosomes that stores genetic information for body design and functioning.

Asexual Reproduction

In asexual reproduction, a single organism gives rise to a new organism. No gametes are formed and no fertilisation takes place. Offspring are usually very similar to the parent, though small variations can arise during DNA copying.

Asexual reproduction is common in unicellular organisms and in simple multicellular organisms. It is faster than sexual reproduction. It does not need a mate.

Fission: Binary and Multiple

Fission means splitting of the parent body into new organisms.

1. Binary fission. The parent cell splits into two equal halves. Each half grows into a new individual. Seen in Amoeba and bacteria. In Amoeba, nuclear division happens first. Then the cytoplasm divides. In Leishmania, which has a whip-like flagellum, binary fission happens in a definite plane.

2. Multiple fission. The parent cell divides into many daughter cells at the same time. Seen in Plasmodium (malarial parasite). Inside a cyst, the nucleus divides many times. Then many daughter cells are formed. They are released when conditions become favourable.

Remember

Binary fission gives 2 daughter cells. Multiple fission gives many daughter cells at once.

Budding, Fragmentation and Regeneration

Budding. A small outgrowth called a bud appears on the parent body. The bud grows and then separates to become a new individual. In yeast, the bud is a small bulge of the cell. It may remain attached for some time and form a chain. In Hydra, a multicellular animal, a bud develops due to repeated cell division at one site and grows into a new Hydra.

Fragmentation. The body of a simple multicellular organism breaks into two or more pieces. Each piece grows into a new organism. Seen in Spirogyra (a filamentous alga). But this does not happen by chance in all organisms. The organism must be simple with cells that can grow independently.

Regeneration. If the body of some animals is cut into pieces, each piece can grow into a complete organism. This is called regeneration. Seen in Planaria and Hydra. It happens with the help of specialised cells. These cells divide and form new tissues. Regeneration is not the same as reproduction in all cases. It is a method of reproduction only in some simple animals.

Vegetative Propagation and Spore Formation

Vegetative propagation in plants. New plants can grow from vegetative parts like roots, stems and leaves. This is called vegetative propagation.

Examples: Potato tubers have buds called eyes. Each eye can grow into a new plant. Bryophyllum has buds in the notches of its leaves. Each bud can form a new plant. Money plant and rose can be grown by stem cuttings. Vegetative propagation can be natural or artificial (cutting, layering, grafting).

Advantages: Plants grown this way preserve most parental features, can bear flowers and fruits earlier, and do not need seeds. A disadvantage is low variation, which can reduce adaptability to environmental change.

Tissue culture. In the laboratory, small pieces of plant tissue are grown in a sterile medium with nutrients and hormones. They form callus and then plantlets. Many plantlets can be grown this way.

Spore formation. Seen in Rhizopus (bread mould), ferns and mosses. The organism forms spores inside a sporangium. Spores are covered by thick walls. They survive harsh conditions. When conditions are good, each spore grows into a new organism by germinating.

Sexual Reproduction in Flowering Plants

Sexual reproduction involves two parents. Male and female gametes fuse to form a zygote. This process is called fertilisation.

Structure of a flower. Flower is the reproductive part of a plant. A typical flower has four whorls: sepals, petals, stamens and carpels.

- **Stamen** is the male part. It has anther and filament. Anther produces pollen grains. Pollen grains contain male gametes.
- **Carpel (pistil)** is the female part. It has stigma, style and ovary. Ovary contains ovules. Each ovule has an egg cell (female gamete).

Flowers can be unisexual (only stamen or only carpel, as in papaya, watermelon) or bisexual (both stamen and carpel, as in hibiscus, mustard).

Pollination. Transfer of pollen from anther to stigma. If pollen falls on the stigma of the same flower, it is self-pollination. If pollen goes to another flower, it is cross-pollination. Cross-pollination is done by wind, water, insects and animals.

Fertilisation in plants. After pollination, a pollen grain germinates on the stigma. It forms a pollen tube that grows through the style and reaches an ovule in the ovary. A male gamete fuses with the egg to form a zygote.

After fertilisation, the zygote divides and forms an embryo. The ovule becomes a seed. The ovary becomes a fruit. Petals and other parts fall off. Seed contains the future plant. It germinates under suitable conditions.

Sexual Reproduction in Human Beings

Human beings reproduce by sexual reproduction. Both male and female have specialised reproductive systems. Gametes are formed: sperms in males, ova in females. Fertilisation is internal (inside the female body).

Puberty is the period during adolescence when reproductive organs mature. Its timing varies from person to person. Hormones control these changes; testosterone and oestrogen are important examples.

Male Reproductive System

The male reproductive system consists of:

- **Testes:** pair of oval organs inside the scrotum. They lie outside the abdominal cavity because sperm formation needs a lower temperature. Testes make sperms and the hormone testosterone.
- **Scrotum:** sac that holds the testes and keeps them cooler than body temperature.
- **Vas deferens:** tube that carries sperms from testes to urethra.
- **Seminal vesicle and prostate gland:** they add their secretions to sperms. This fluid is called semen. It provides nutrition and helps sperms to move.
- **Urethra:** common tube for semen and urine. It opens outside through the penis.
- **Penis:** organ that transfers sperms into the vagina during intercourse.

Sperms are tiny. Each sperm has a head, middle piece and a tail. The tail helps it to swim.

Female Reproductive System and Menstrual Cycle

The female reproductive system consists of:

- **Ovaries:** pair of organs in the abdominal cavity. They produce ova (eggs) and hormones oestrogen and progesterone. One egg matures in one ovary every month.
- **Oviduct (fallopian tube):** tube with funnel-like opening near the ovary. It carries the egg from ovary to uterus. Fertilisation happens here.
- **Uterus (womb):** bag-like organ where the embryo develops. Its wall is thick and richly supplied with blood vessels.
- **Vagina:** tube that connects uterus to outside. Sperms enter through vagina.

Menstrual cycle. Usually, an ovary releases an egg during each menstrual cycle; this is called ovulation. In a typical 28-day cycle it may occur near the middle, but cycle length and timing vary.

The uterus prepares itself every month to receive a fertilised egg. Its lining becomes thick and spongy with many blood vessels. This is to nourish the embryo if fertilisation happens.

If the egg is not fertilised, the thickened uterine lining breaks down and leaves the body as blood and tissue. This is menstruation. The duration and interval vary among individuals and from cycle to cycle.

If the egg is fertilised by a sperm, a zygote forms in the oviduct. It divides to form an embryo, which later implants in the uterine wall. Menstruation usually stops during pregnancy, although a missed period alone does not confirm pregnancy.

The embryo gets nutrition from the mother through a special tissue called placenta. Placenta is a disc embedded in the uterine wall. It has villi with blood spaces. Glucose, oxygen and other substances pass from mother to embryo. Waste goes from embryo to mother. The embryo develops for about nine months and is then born as a child.

Reproductive health. Early pregnancy and sexually transmitted infections (STIs) are health concerns. Contraceptive methods include barriers such as condoms, hormonal oral pills, intrauterine devices such as Copper-T, and surgical methods such as vasectomy and tubectomy. Condoms reduce the risk of HIV and several other STIs when used correctly, though no method eliminates risk completely.

Summary

- DNA copying is the basic event in reproduction and it brings small variations.
- Asexual reproduction needs one parent. Types: binary fission (Amoeba), multiple fission (Plasmodium), budding (yeast, Hydra), fragmentation (Spirogyra), regeneration (Planaria), vegetative propagation (potato, Bryophyllum) and spore formation (Rhizopus).
- Flower is the reproductive part of a plant. Stamen is male, carpel is female.
- Pollination is transfer of pollen to stigma. Fertilisation is fusion of male and female gametes to form a zygote.
- After fertilisation, ovule becomes seed and ovary becomes fruit.
- Male system: testes make sperms and testosterone. Semen is sperms plus gland secretions.
- Female system: ovaries make eggs. Fertilisation happens in oviduct. Embryo develops in uterus.
- Menstrual cycle is a 28-day cycle. If egg is not fertilised, uterine lining breaks and menstruation occurs for 2 to 8 days.
- Placenta connects mother and embryo for nutrition, gas exchange and waste removal.

- Contraception methods help to avoid unwanted pregnancy and infections.

Practice Questions

1. Why is DNA copying an essential part of the process of reproduction?

2. Differentiate between binary fission in Amoeba and budding in yeast with diagrams.

3. What is vegetative propagation? Name two plants where leaf buds can form new plants. Give one advantage and one disadvantage.

4. Draw a labelled diagram of the longitudinal section of a flower showing its reproductive parts.

5. Describe the male and female reproductive systems in humans with labelled diagrams.

6. What is the menstrual cycle? What happens if the egg is not fertilised? What happens if it is fertilised?